

AKHILA BODEPUDI

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PROFESSIONAL SUMMARY

Python Full Stack Developer with nearly **three years of experience** designing and delivering backend services, responsive frontends, and AI-enabled features for enterprise applications. Core expertise includes **Python, Django, FastAPI, Flask, Angular, React.js, REST APIs, microservices, and SQL**, along with **machine learning, NLP, OpenAI APIs, LangChain, and MLflow** for workflow automation and data-driven insights. Skilled in deploying **containerized applications on AWS and Azure** using **Docker, Kubernetes, and CI/CD pipelines** while collaborating with cross-functional Agile teams.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL

Backend Development: Python, Django, Django REST Framework, FastAPI, Flask, REST APIs, Microservices

Frontend Development: Angular, React.js, Redux, TypeScript, HTML5, CSS3, Bootstrap

AI / Machine Learning: Machine Learning, NLP, Deep Learning, TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, OpenAI APIs, LangChain

Generative AI: LLM Integration, RAG Architecture, Vector Databases, Embeddings, AI Assistants

Data Processing: Pandas, NumPy, PySpark, Apache Spark

MLOps: MLflow, Model Deployment, Model Monitoring, Feature Engineering

Cloud Platforms: AWS, Azure, Azure ML, Azure Functions, Azure Blob Storage, AKS, EKS

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Redis, DynamoDB

DevOps & Containers: Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD

Messaging & Streaming: Apache Kafka, RabbitMQ

Version Control: Git, GitHub, GitLab, Bitbucket

Testing: PyTest, Selenium, Postman, API Testing

Methodologies: Agile, Scrum, CI/CD, DevOps, SDLC

PROFESSIONAL EXPERIENCE

Python Full Stack Developer | Cisco Systems, USA

Jan 2026 – Present

- Designed and developed scalable backend services using **Python, FastAPI, Django REST Framework, and Microservices** architecture, supporting enterprise network management applications handling **10K+ API requests per day**.
- Developed responsive web applications using **Angular, TypeScript, HTML5, CSS3, and Angular Material**, enhancing network monitoring dashboards and improving operational visibility for multiple internal engineering teams.
- Designed and optimized **RESTful APIs** integrating frontend applications, databases, and external network management systems, improving **API response time by 20%** through query tuning and backend optimization.
- Implemented **AI-driven automation** capabilities using **Python ML libraries, NLP techniques, and LLM-based solutions** to automate incident categorization and reduce manual troubleshooting efforts by approximately **15%**.
- Built data processing workflows using **Python, Pandas, NumPy, and SQL**, processing **millions of operational data** records to support analytics, reporting, and intelligent decision-making.
- Developed containerized applications using **Docker and Kubernetes** and deployed solutions across **AWS cloud environments**, improving application scalability and reducing deployment-related issues by approximately **25%**.
- Automated **CI/CD workflows** using **GitHub Actions, Jenkins, and testing frameworks**, reducing manual deployment activities by **30%** and improving release efficiency across development environments.
- Performed **code reviews, debugging, and performance optimization** activities, improving application stability and reducing recurring production issues by approximately **15%**.

Python Full Stack Developer | Hexaware Technologies, India

Jun 2022 – Aug 2024

- Developed enterprise applications using **Python, Django, Flask, FastAPI, React.js, JavaScript, and TypeScript**, delivering scalable solutions supporting **5K+ business transactions daily**.
- Designed and implemented **REST APIs and microservices** for customer management, workflow automation, and third-party integrations, improving application scalability and reducing **processing time by approximately 20%**.
- Built reusable frontend components using **React.js, Redux, HTML5, CSS3, and Bootstrap**, improving application performance and reducing **page load times by nearly 18%**.
- Developed **Machine Learning and NLP-based solutions** using **Python, Scikit-learn, TensorFlow, and PyTorch** to automate text classification, sentiment analysis, and document processing workflows.

- Managed **ML lifecycle activities using MLflow**, including experiment tracking, model versioning, and deployment monitoring for multiple machine learning models, improving model management efficiency.
- Created **Docker-based development environments** and deployed applications using **Kubernetes on Azure**, reducing **environment setup time by approximately 25%**.
- Developed data transformation pipelines using **Python, Pandas, SQL, and cloud services**, processing **large datasets** and improving data preparation efficiency for analytics and AI workloads.
- Implemented **automated testing using PyTest and API testing frameworks**, increasing test coverage and reducing production defects by **approximately 20%**.
- Collaborated with **Agile teams** to deliver sprint-based enhancements, contributing to the successful delivery of multiple enterprise application releases within planned timelines.

ACADEMIC PROJECTS

AI-Powered Customer Support Assistant

Description: Developed an AI-driven customer support application using Python, NLP, and modern web technologies to automate customer query handling and provide intelligent responses through a conversational interface.

Technologies: Python, FastAPI, React.js, NLP, Machine Learning, OpenAI API, LangChain, PostgreSQL, Docker

- Developed backend services using Python and FastAPI to process customer requests and integrate AI-based response generation workflows.
- Built responsive user interfaces using React.js, JavaScript, HTML5, and CSS3 for chatbot interaction and support dashboards.
- Implemented NLP techniques for text classification, intent detection, and automated query processing.
- Integrated AI models using OpenAI APIs and LangChain to improve response accuracy and knowledge retrieval.
- Containerized the application using Docker and deployed services in a cloud-based environment.

AgroNity - Full-Stack Agriculture Marketplace & ML Platform

Description: Built and deployed a production Django web app that combines an online crop marketplace with a machine learning recommendation engine, giving farmers, customers, and admins one platform to buy, sell, and make data backed crop decisions.

Technologies: Python, Django, XGBoost, scikit-learn, NumPy, PostgreSQL, SQLite, HTML5, CSS3, Bootstrap 5, Leaflet.js, Vercel

- Designed and built the backend in Django, including custom user models, role-based auth (Farmer/Customer/Admin), and an admin approval workflow.
- Built the marketplace end to end (crop listing, editing, cart, and checkout) with ownership checks so users can only edit their own listings.
- Trained an XGBoost model on soil and weather data (N, P, K, temperature, humidity, pH, rainfall) that hit 99.5% accuracy for crop recommendations, and wired it into the Django app for live predictions.
- Added a Leaflet/OpenStreetMap integration so farmers can locate nearby fertilizer shops and soil-testing centers.
- Handled deployment myself: set up Vercel for serverless hosting, migrated from SQLite to Neon PostgreSQL for production, and configured WhiteNoise for static file delivery.

EDUCATION:

University of Maryland, Baltimore County (UMBC)

Master of Science (M.S.) in Computer Science

Baltimore, Maryland, USA

August 2024 - May 2026

CERTIFICATIONS:

- Microsoft – Microsoft Certified: Azure AI Engineer Associate
- Google Cloud – Google Cloud Certified Associate Cloud Engineer
- Google Cloud Learning Services – Associate Cloud Engineer Track
- TATA Group – Data Visualization: Empowering Business with Effective Insights
- NPTEL – Cloud Computing